Multiple IV Infusions & Multiple Oral Practice 1

Module 6 - Multiple Dosings · Daily practice sheet · No date

Name _____ Date _____

1a 1. An 800-mg dose of a medication was administered as an IV infusion over a period of 3 hours to an 80-kg patient. Six hours after the start of the first infusion, a second of 800-mg dose was infused, again over 3 hours. The drug has an apparent volume of distribution that is 25% of body weight and a total body clearance is 4.62 L/hr. a. What is the plasma drug concentration at the end of the first infusion?

1b 1. An 800-mg dose of a medication was administered as an IV infusion over a period of 3 hours to an 80-kg patient. Six hours after the start of the first infusion, a second of 800-mg dose was infused, again over 3 hours. The drug has an apparent volume of distribution that is 25% of body weight and a total body clearance is 4.62 L/hr. b. What is the plasma drug concentration 6 hours after the cessation of the second infusion?

2a 2. An adult male (80 kg) was given 500 mg of an antibiotic orally every 12 hours for 2 weeks. The literature reports that the antibiotic is about 90% bioavailable and has a V_D of 0.24 L/kg. The elimination half-life is about 7.3 hours and the absorption half-life is about 75 minutes. Calculate a. Time required to reach the maximum concentration of the first dose.

2b 2. An adult male (80 kg) was given 500 mg of an antibiotic orally every 12 hours for 2 weeks. The literature reports that the antibiotic is about 90% bioavailable and has a V_D of 0.24 L/kg. The elimination half-life is about 7.3 hours and the absorption half-life is about 75 minutes. Calculate b. Maximum plasma drug concentration of the first dose.

2c 2. An adult male (80 kg) was given 500 mg of an antibiotic orally every 12 hours for 2 weeks. The literature reports that the antibiotic is about 90% bioavailable and has a V_D of 0.24 L/kg. The elimination half-life is about 7.3 hours and the absorption half-life is about 75 minutes. Calculate c. Minimum plasma drug concentration of the first dose.

2d 2. An adult male (80 kg) was given 500 mg of an antibiotic orally every 12 hours for 2 weeks. The literature reports that the antibiotic is about 90% bioavailable and has a V_D of 0.24 L/kg. The elimination half-life is about 7.3 hours and the absorption half-life is about 75 minutes. Calculate d. Area under the curve of the first dose.

2e 2. An adult male (80 kg) was given 500 mg of an antibiotic orally every 12 hours for 2 weeks. The literature reports that the antibiotic is about 90% bioavailable and has a V_D of 0.24 L/kg. The elimination half-life is about 7.3 hours and the absorption half-life is about 75 minutes. Calculate e. Time required to reach the maximum concentration at steady state.

2f 2. An adult male (80 kg) was given 500 mg of an antibiotic orally every 12 hours for 2 weeks. The literature reports that the antibiotic is about 90% bioavailable and has a V_D of 0.24 L/kg. The elimination half-life is about 7.3 hours and the absorption half-life is about 75 minutes. Calculate f. Maximum plasma drug concentration at steady state.

2g 2. An adult male (80 kg) was given 500 mg of an antibiotic orally every 12 hours for 2 weeks. The literature reports that the antibiotic is about 90% bioavailable and has a V_D of 0.24 L/kg. The elimination half-life is about 7.3 hours and the absorption half-life is about 75 minutes. Calculate g. Minimum plasma drug concentration at steady state.

2h 2. An adult male (80 kg) was given 500 mg of an antibiotic orally every 12 hours for 2 weeks. The literature reports that the antibiotic is about 90% bioavailable and has a V_D of 0.24 L/kg. The elimination half-life is about 7.3 hours and the absorption half-life is about 75 minutes. Calculate h. Average plasma drug concentration at steady state.

2i 2. An adult male (80 kg) was given 500 mg of an antibiotic orally every 12 hours for 2 weeks. The literature reports that the antibiotic is about 90% bioavailable and has a V_D of 0.24 L/kg. The elimination half-life is about 7.3 hours and the absorption half-life is about 75 minutes. Calculate i. Dose required to reach an average steady state conc of 25 mg/L.

2j 2. An adult male (80 kg) was given 500 mg of an antibiotic orally every 12 hours for 2 weeks. The literature reports that the antibiotic is about 90% bioavailable and has a V_D of 0.24 L/kg. The elimination half-life is about 7.3 hours and the absorption half-life is about 75 minutes. Calculate j. Average plasma drug concentration at steady state if the dose were increased to 750 mg.

2k 2. An adult male (80 kg) was given 500 mg of an antibiotic orally every 12 hours for 2 weeks. The literature reports that the antibiotic is about 90% bioavailable and has a V_D of 0.24 L/kg. The elimination half-life is about 7.3 hours and the absorption half-life is about 75 minutes. Calculate k. Average plasma drug concentration at steady state if the dosing interval was decreased to 8 hours (dose 500 mg).